EVENT HORIZON

Math case study · Deepreel Studio · November 2026

Event Horizon is a 6x5, 50-line slot built around a **persistent charge grid**. Wins through a wild charge their cells in the colour of a wild faction (Nova or Void); the charge stays between spins; a **black hole** cashes the whole grid out, multiplied by the wilds of the faction holding the most charge. A rare **Quasar** wild multiplies its side's total instead of adding to it.

Playable demo deepreel-studio.github.io/event-horizon

KEY FIGURES · 96% VERSION

Metric	Value
Simulated RTP (640M continuous spins)	96.03% ± 0.05%
RTP split	lines 32.0% · black holes 4.9% · free spins 49.1% · features 10.3%
Hit rate · standard deviation	37.4% · 21.1x per spin
Black hole (base game)	1 in 715, average 35x
Free spins	1 in 235, average 115x, minimum 6x
Mode medians (same average value)	Orbit 75x · Nebula 60x · Singularity 37x
Big Bang (5+ BONUS, hidden bonus)	1 in ~195,000, average 345x
Max win 10,000x	about 1 in 2.4M spins

BUYS AND BET MODES · FIXED PRICES, RTP-MATCHED

Product	Price	94%	96%	97%
Free Spins (pick a mode)	120x	94.1%	96.0%	97.1%
Super Free Spins (black hole x3 from the start)	600x	93.8%	96.0%	96.9%
Feature spins: Surge / Alignment / Supernova	35x / 190x / 630x	93.9–94.1%	95.9–96.1%	97.0–97.2%
Bonus Hunt / Super Bonus Hunt (free spins 3x / 11x more often)	x2 / x6 per spin	94.0%	96.0%	96.9–97.0%
Quasar Spins (a Quasar on every spin)	x2.2 per spin	94.0%	96.0%	96.9%

Prices are identical in every RTP version. Each product has its own continuous tuning knob (black hole rate per mode, Quasar rate per feature and bet mode), set by secant search on 8–10M rounds and checked on independent seeds.

CONTROLLING THE PERSISTENT STATE

A visible charge means the next spin is worth more on a charged grid. In early versions the next spin returned 150% with 10–19 charge and 195% with 20–29. Splitting the return by source showed two causes, both fixed:

- Natural features cashed the current grid with large multipliers: they now play on their own grid.
- The black hole was too frequent for the value stored on the grid: it is now rarer and bigger (1 in 600 charged spins).

Grid charge before the spin	Share of spins	Next-spin return (96%)
0–9	71.6%	94.4%
10–19	24.6%	99.4%
20–29	3.5%	105.0%
30–39	0.3%	109.9%

The grid is saved per bet level and per bet mode, and every bonus and feature plays on its own grid, so no stake or mode switch can cash a charged grid at a better price.

WHAT THE AUDIT CHANGED

- **No empty bonus.** 26% of Singularity rounds and 11% of Nebula rounds paid under 1x. Free spins now start with the triggering BONUS symbols as charged cells: minimum 6x, medians up 1–10x.
- **Equal-value modes.** Orbit, Nebula and Singularity return the same average (within 0.1%) with very different volatility.
- **Max win includes same-spin free spins**, capped at 10,000x.
- **Engine parity.** The JavaScript demo engine mirrors the Python simulator; line, black hole and free spin frequencies match within simulation noise.

METHOD

Python + Numba, parallel chains of continuous play (grid, decay counters and charge carry from spin to spin). Free spins and features inside the chains are valued at their simulated averages for variance reduction. Full reel strips, paytables, distributions and per-state return are in the PAR sheet, available on request.

Deepreel Studio · Lotfi Seddiki · khedam.317@gmail.com · Demo: deepreel-studio.github.io/event-horizon